

EyeClick: A Robust Two-Step Eye-Hand Interaction for Text Entry in Augmented Reality Glasses

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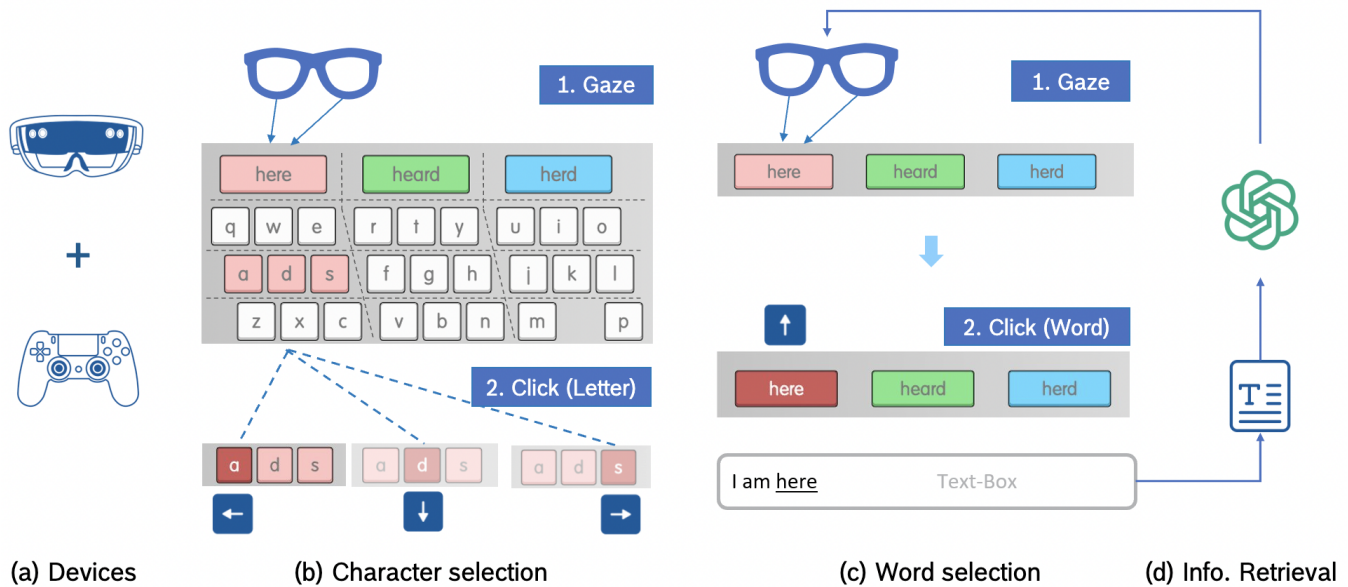


Figure 1: EyeClick (RowType variation). (a) AR Glasses with a handheld controller; (b) Character selection: (1) select a block via gazing, (2) select individual letters via clicking up, down, or right on the controller; (c) Word selection: (1) point to the column via gazing, (2) select the word via clicking up on the controller; (d) Integration with ChatGPT for information retrieval.

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ABSTRACT

Efficient text entry is a crucial aspect of the user experience for augmented reality (AR) head-mounted displays (HMD). Eye-tracking for virtual keyboard interaction is a popular choice for AR text entry, as it is intuitive, privacy-preserving, and socially acceptable. However, as AR HMDs move toward more consumer-friendly form factors and price points, technical constraints necessitate trade-offs that result in a limited field of view (FoV) and reduced eye-tracking accuracy.

To address these challenges, we develop EyeClick, a novel two-step eye-hand interaction method utilizing a modified QWERTY keyboard for AR HMDs. To type, users first employ gaze to select a large area containing multiple characters, then press a button on a handheld controller to choose a single character from the selected area. We demonstrated competitive speed and reduced corrected error rate (CER) with two variants of the EyeClick design: ColType (7.30 WPM) and RowType (9.41 WPM).

We also integrated ChatGPT with the EyeClick keyboard to build an interactive information retrieval AR application. We demonstrate that the EyeClick keyboard can help users efficiently extract information about surroundings by typing to ChatGPT in mixed reality.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality**; **Text input**.

KEYWORDS

text entry, text input, augmented reality, gaze detection, eye-tracking

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1 INTRODUCTION

An augmented reality (AR) head-mounted display (HMD) is a wearable device that overlays digital information onto a user's view of the physical environment. Industry-leading AR HMDs such as HoloLens 2 and MagicLeap 2 are widely used in professional settings including healthcare [5], design, and manufacturing [8] for applications in immersive training [4], remote guidance, and data visualization. However, their cost and form factor (\$3,000+) have limited their wide adoption in the consumer market. Google [6], Nreal [13], and Vuzix [20] now offer affordable (\$200 to \$1,000) everyday AR glasses for applications in sports, social networking, tourism, messaging [17], navigation [18], and real-time translation [10]. These consumer-oriented AR glasses have sleeker form factors, and are growing more popular. However, lower cost can limit the capabilities of these AR glasses, imposing constraints in field-of-view (FoV) and eye-tracking accuracy [1, 12, 19]. This could lead to challenges in designing natural and robust user interactions in these affordable everyday AR glasses.

Efficient text entry, a core AR functionality, is essential for user experience in both professional and consumer AR markets. There are two major input modalities for text entry in AR devices: (1) Automatic speech recognition (ASR), (2) Virtual keyboard typing. ASR is more efficient, reaching up to 153 words per minute (WPM) in English [16], making it ideal for AR text entry. However, it is prone to errors in noisy environments, and more importantly, it is not private in social situations such as parties, social gatherings, public events, and of course, in libraries.

Therefore, a virtual keyboard typing technology is necessary to complement ASR in most AR HMDs and AR glasses. Yet, most virtual keyboard typing experiences suffer from low speed and can be awkward to use. Most of these, particularly gesture-controlled virtual keyboards, do not provide sufficient privacy. Moreover, they rarely justify their advantage over ASR in environments with rich human interactions.

In this work, we developed a natural, socially acceptable, and robust virtual keyboard AR text entry technology that is robust against variations in keyboard sizes and the quality of eye-tracking. By creating a private two-step eye-hand typing process combined with a novel virtual keyboard design called EyeClick (Figure 1), we were able to compensate for (1) AR displays with a smaller FoV, and (2) low accuracy in eye-tracking. Integrating with Large Language Models (LLM) such as ChatGPT, we demonstrate efficient in-context information retrieval. This EyeClick keyboard can be adapted to both professional AR HMDs and consumer-facing everyday AR glasses alike while maintaining typing efficiency and accuracy.

2 DESIGN OF EYECCLICK

EyeClick aims at providing efficient text entry approach for consumer-oriented, low-cost, everyday AR glasses. We design EyeClick with the following requirements: (1) it should be robust to various FoVs and eye-tracking techniques; (2) interactions with the virtual keyboard should be socially acceptable and privacy-preserving.

In this section, we first expand the above design requirements and introduce the design justifications of EyeClick; next, we demonstrate the system design of EyeClick, including the interface, interaction approach, and word suggestion mechanisms.

2.1 Design Rationale

Social Acceptance and Privacy. Literature on wearable devices shows that, in addition to usability and cost, social influence also affects users' adoption of new technology [7, 11, 14]. However, most previous work has not considered situations in which the user is in a public area. A short survey by Ahn et al. showed that people are more willing to perform text entry on a smartwatch than on a touchpad on glasses or through mid-air gestures when in the presence of others [2]. Although ASR offers an efficient alternative that can be easily implemented on AR glasses, speaking text aloud may raise privacy concerns for users. A private, yet efficient, virtual keyboard implementation should complement ASR in human-rich environments.

Robustness. Robustness of gaze-assisted text entry design on AR HMDs can be affected by two factors: tracking accuracy and field of view (FoV). The FoV directly influences the maximum keyboard size that AR glasses can display. Previous research has observed inconsistent text entry performance on AR and VR HMDs, even when using similar interaction designs [9, 21]. Consumer-oriented AR glasses prioritize reducing cost and weight to promote broader adoption. As a result, they tend to have a smaller FoV compared to professional AR HMDs and general VR HMDs. Consequently, there is a need for efficient text entry methods that can accommodate these constraints.

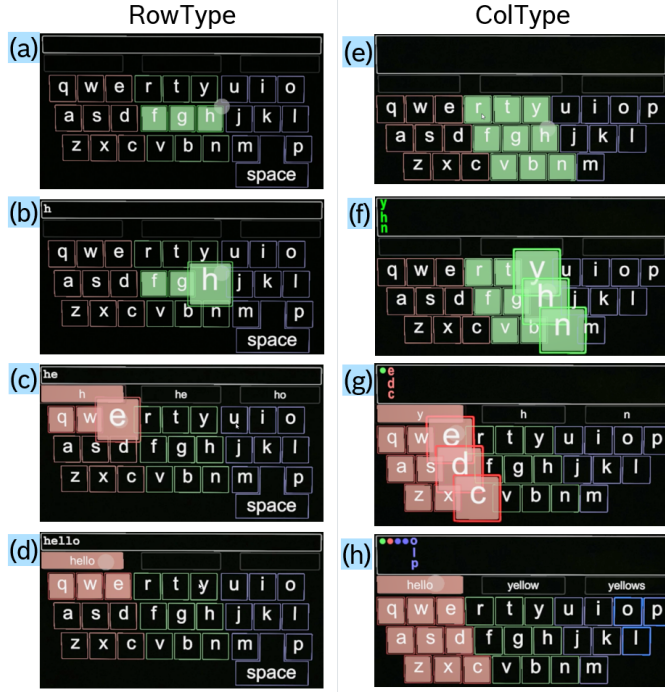


Figure 2: Typing "hello". Left: RowType. (a) eye gaze to select a three-letter block, highlighted as green, (b) button click "right" to select a letter, highlighted and popped out, (c) eye gaze plus button click to select the next letter, (d) eye gaze to select the column of word suggestion, then button click "up" to select the word. Right: ColType, (e) eye gaze to select a nine-letter block, highlighted as green, (f) button click "right" to select the column, (g) same as (c), (h) same as (d).

2.2 Typing Interface and Interaction

EyeClick is a two-step eye-hand interaction text entry method utilizing a modified QWERTY keyboard for AR HMDs. A user first employs gaze input to select a large area containing multiple characters, then presses a button on a handheld controller to choose a single character from the selected area. The controller allows inputs in four discrete directions, so it can be easily controlled by simple controllers or other gestures. In this paper, we use an Xbox controller to demonstrate the prototype. As previous work [3, 15] pointed out, the performance of eye gaze depends significantly on eye-tracking accuracy, and it may vary among users. Therefore, we propose two variations of EyeClick: RowType and ColType.

Figure 2 shows the interface of EyeClick, containing a text input box, a word suggestion list, and a modified QWERTY keyboard. The keyboard columns are color-coded red, green, and blue and groups of keys are highlighted when a user looks at the area.

RowType breaks the keyboard into a 3 x 3 grid. The user first employs eye gaze to select a coarse section that contains three letters, such as "QWE", "FGH", and "M[space]P". Then the user selects the single letter by tapping left, down, or right on the controller. **ColType** expands the coarse section to nine characters and the fine section to three characters. Users select the column instead of

individual characters. The selected column tri-grams are decoded by a language model that generates word suggestions.

2.3 Word Suggestions

Language models (LMs) and statistical decoding have long been integrated into text entry systems. To enhance text entry, we incorporated a language model to provide word suggestions. Specifically, we used a trigram model, which estimates the probability of the next word in a sequence based on the two preceding words. The Maximum Likelihood Estimator (MLE) approach is employed to optimize the prediction of the most probable word in the given context.

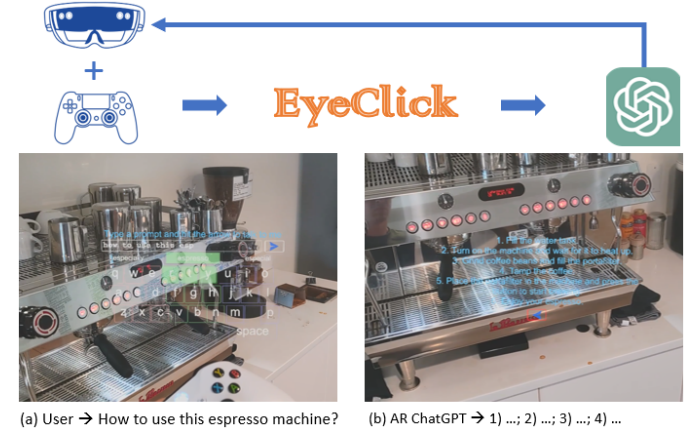


Figure 3: EyeClick Information Retrieval Demo in AR: A user used EyeClick to ask ChatGPT how to use the Espresso machine in AR. The user can use AR glasses to request general information and procedures from ChatGPT about his/her surroundings interactively. This application often requires efficient text input via a privacy-preserving virtual keyboard, i.e., library, social gatherings, and manufacturing floor.

2.4 Demo: EyeClick with ChatGPT

After implementing prototypes of EyeClick, we were looking for a killer application to demonstrate the usefulness of EyeClick in AR. We integrated the EyeClick virtual keyboard together with ChatGPT to demonstrate how to request information from ChatGPT from the surroundings in AR. AR pass-through allows the user to observe their surroundings while asking ChatGPT questions via texting. Such an interactive interface is particularly useful to retrieve information and procedures, for example, how to use a coffee machine in the office (see Figure 3); how to bake a cake; what a book in the library is about; etc. EyeClick is both robust and private for the user to interact with ChatGPT in AR. Such applications combined with EyeClick can greatly improve user experiences for people who are using AR for both everyday tasks and professional work.

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